Amit Singh Rajawat

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Technical Skills

Programming Languages: C/C++, Python, JavaScript, Shell, MySQL
Developer Tools & Frameworks: Git, GitHub, Linux, Docker,ONNX, AWS, CI/CD, MLOps, Prometheus, Grafana
Libraries: PyTorch, TensorFlow, Scikit-learn,Flask, OpenCV, Transformers, MXNet, LLMs, FastAPI

Experience

Software Engineer — Jun 2024 – Present
Kloudspot Inc. — *Bengaluru, India*

- Developed **Face Recognition Algorithm** in **C++** for **1:1 Verification** and **1:N Identification**, with **FRS Analysis & Registration Tools** in Python using clustering techniques for multi-pose registration, real-time tracking, and visualization.
- Implemented a custom **NVIDIA DeepStream** parser for **face detection** and **landmark extraction**, generating **optimized engine files** for **NVIDIA platforms**, achieving **30 fps** in **real-time facial analysis** on **4K video streams**.
- Engineered **scalable data pipelines** for **real-time facial recognition**, with **CI/CD** integration for **automated deployment**, collaborating with teams to define **performance metrics**, improving **system efficiency** by **15%**.
- Trained models across architectures (ResNet, ResNet+CBAM, ResNet+attention), using hyperparameter tuning, knowledge distillation, and reinforcement-learning techniques to improve accuracy by **13%**.
- Designed **C++** algorithm for **Real-Time Multi-Camera Panoramic Stitching** reducing screens by at least **50%** using **ORB** incorporating seam exposure, composing, blending for seamless stitching with real-time camera synchronization.
- Developed LiDAR-based airplane ID pipeline leveraging **PointPillars** (**85% acc**) **PV-RCNN** (**90% acc**) on airplane dataset; from preprocess to temp-matching vs live LiDAR streams, optimized inference for RT deployment.
- Trained **action-recognition model** from scratch using **MViT** for retail actions (pick/hold/drop/remove/return), achieving **97.46%** accuracy; curated dataset/augmentation, end-to-end training, RT inference for in-store monitoring.
- Engineered high-performance C++ CV inference framework (**C++**, **OpenVINO**, **DLStreamer**) with **gRPC**-based GStreamer **RTSP** server, opt. client–server comm pre/post-processing for large-scale RT inference; imp. speed by **9%**.

Machine Learning Intern (LOR) — Dec 2023 – Jan 2024
Indian Space Research Organisation (ISRO) — *Bengaluru, India*

- Designed **vision-based satellite pose estimation system** using **HRNet** algorithm, iterative hyperparameter tuning, real-time processing optimizations; achieved **90%** landmark-detection accuracy across diverse imaging conditions.
- Integrated **deep landmark regression** with **nonlinear pose refinement**, reducing pose error by **15%** through extensive model training and validation.

Projects

EchoNeuron AI Engine | *Python, FastAPI, LangChain, ChatGroq, Tavily, JavaScript, Render*

- Built an **unified assistant** with **live web lookup**, **DuckDuckGo search**, **YouTube** and **PDF summarization** (**40%** content reduction), **tutoring**, and **quiz generation** using **FastAPI**, **LangChain**, and **ChatGroq**.
- Packaged **GUI** in **Docker** with **secure CI/CD pipeline** on **Render**, reducing deployment time by **40%**.

StoryCraftAI | *Python, TensorFlow, Keras, NLP, Pandas, Hugging Face, CI/CD*

- Implemented and enhanced **deep-learning models** for **story generation** using **advanced NLP techniques**, improving performance across **GRU** (+13.46%), **LSTM** (+33.53%), **Bi-LSTM** (+25.32%) and **Bi-GRU** (+9.91%).
- Built an **interactive Flask** and deployed via **CI/CD** on **Hugging Face Spaces** for **real-time story generation**.

Education

Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal — Aug 2020 – May 2024
B.Tech in Information Technology (Artificial Intelligence & Robotics) — *CGPA: **8.65***

Publications

- "**Categorization of IRIS Flower using Different Machine Learning Strategies**" [paper]
- "**Design and Analysis of Programmed Face Monitoring System**" [paper]
- "**Recognition of Parkinson's Ailment by Using Various Machine Learning Procedures**"[paper]
- **On Heart Disease** (Communicated in Journal).

ACHIEVEMENTS

- **LeetCode:** 1100+ problems solved, 1765+ rating (Profile), **GeeksforGeeks:** 320+ problems solved. (Profile)
- Selected for **IIT Madras Pravartak Undergraduate Fellowship 2023–24** (one of 72 nationwide). (Link)
- Led development of **Novel Engaging Course (NEC) Portal**, used by 5,000+ students. (Portal)
- Organized 3-day hackathon at **MITS** as core organizing committee member.
- Selected for the **Amazon ML Summer School**, implementing **machine-learning algorithms** in **Python** to achieve **over 95% accuracy** and building **sequence-modeling** and **Spark-based pipelines** for **high-throughput, low-latency inference**.